Md. Mahbubur Rahman

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Education

BSc in Computer Science and Engineering — Feb 2022 – Oct 2026*
United International University — Dhaka, Bangladesh

Technical Skills

Languages: C, C++, Python, JavaScript, PHP
Frontend: React, React Native, HTML, CSS, Tailwind CSS
Backend: Node.js, Express.js
Databases: MySQL, PostgreSQL, MongoDB, Redis
ORM & ODM: Prisma, Mongoose
Tools: Git, GitHub, Postman
Concepts: Data Structures & Algorithms, REST APIs, OOP.

Projects

Ziplify — A URL Shortener
Tech Stack: React, Node.js, Express.js, JavaScript, PostgreSQL, Prisma, Redis, Vercel, Render
Source Code: Backend | Frontend | **Live:** ziplify.vercel.app

- Architected a Base62-encoded URL shortener with a Redis cache-aside pattern, reducing redirect latency by 39% (286ms → 176ms), validated through repeated load testing.
- Implemented Redis-backed distributed rate limiting across independently deployed frontend (Vercel) and backend (Render) services to ensure consistent limits across instances.
- Deployed backend with PostgreSQL (Neon) via Prisma ORM, using atomic sequence-based ID generation to avoid write contention without manual collision handling.
- Resolved production issues spanning CORS configuration, cross-service routing via edge rewrites, and TypeScript runtime deployment on Render.

Wanderlust — Full-Stack Property Rental Platform
Tech Stack: Node.js, Express.js, MongoDB, Mongoose, Cloudinary, JavaScript, Bootstrap
Source Code: Code | **Live:** wanderlust-3r2u.onrender.com

- Engineered a full-stack rental platform with secure authentication, role-based authorization, and RESTful APIs for managing listings, reviews, and users.
- Designed scalable data models using MongoDB (Mongoose) with relational referencing for listings, users, and reviews, ensuring data consistency and efficient querying.
- Integrated geolocation and cloud-based media storage (Cloudinary) to support map-based property discovery and optimized image handling for improved user experience.

SatheCholo — Ride-Sharing & Carpooling Platform
Tech Stack: Node.js, Express.js, PostgreSQL, WebSocket, Firebase, OpenStreetMap

- Building a bilingual (English & Bangla) ride-sharing platform supporting both intra-city rides and inter-city carpooling.
- Implemented a real-time ride-matching and chat system using WebSockets, reducing matching latency.
- Implementing scalable backend architecture with Express and event-driven communication.
- Added safety features including SOS emergency support, and AI-based harassment detection.

DirectEdge — Agricultural Supply Chain Management System
Tech Stack: Python, PHP, JavaScript, MySQL
Source Code: Code

- Developed a 3-sided platform connecting farmers, distributors, and retailers to streamline supply chain operations.
- Implemented real-time inventory tracking and demand forecasting using machine learning techniques.
- Improved operational efficiency by reducing manual coordination and enabling data-driven decision making.

Competitive Programming

- Solved 335+ problems on **Codeforces**,. Codeforces max rating: 950 (Newbie).
- Participated in 40+ contests on **Codeforces** and **VJudge**.

- Participated in inter-university programming contests including **UIU IUPC**, **UIU Coders Combat 4.0**, **UIU Coders Combat 5.0**.

Achievements

- **First Runner-Up** at the UIU CSE Project Show (Summer 2025) for the project "DirectEdge".
- **9th Runner up** in UIU Coders Combat 5.0.
- Academic Scholarship Recipient.

Experience

Personal Projects & Self-Driven Development

- Built and deployed multiple full-stack applications using modern technologies.
- Followed clean code practices, version control, and scalable system design principles.